

JONGHO PARK

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PROFILE

Machine Learning Engineer and Master's graduate in Data Science at the University of Melbourne with experience in machine learning, data analysis, and AI-driven solutions. Skilled in Python, R and SQL, building data pipelines, analytical models, and dashboards on cloud platforms such as AWS and BigQuery. Experienced in translating business problems into scalable data science solutions

RELEVANT EXPERIENCE

Machine Learning Engineer

Jul, 2025 - Present

ResearchQ, Melbourne, Australia

- Developed an LLM-powered research platform serving 2.7K+ users, transforming scientific documents into interactive learning experiences using retrieval-augmented generation (RAG) pipelines
- Built and optimised a RAG system combining semantic search, hierarchical chunking, and context assembly, improving retrieval relevance by ~20–30% offline evaluation and reducing irrelevant context injection, leading to more consistent and accurate LLM responses
- Optimised RAG retrieval architecture using key-value caching and vector indexing, reducing end-to-end inference latency to ~400ms for real-time production workloads
- Designed a data-driven segmentation pipeline based on 10M+ interaction events, enabling identification of key user groups (students, researchers, professionals) and improving content targeting efficiency (+20% proxy behavioural lift metric)
- Designed and evaluated LLM prompting strategies using few-shot prompting and structured reasoning techniques to improve response quality and consistency
- Built an end-to-end evaluation framework for LLM lifecycle management, including offline evaluation sets, prompt benchmarking, and model comparison across iterations
- Evaluated multiple open-source and commercial LLMs using qualitative and quantitative metrics to benchmark response quality, consistency, and retrieval effectiveness
- Deployed and served production ML/RAG APIs using Cloudflare Workers and cloud infrastructure, enabling scalable low-latency inference serving

Signals Intelligence Sergeant

Jan, 2024 - Jul, 2025

Information Battalion, Republic of Korea Army, South Korea

- Led a team of 10+ personnel to improve intelligence reporting workflows by redesigning data collection, validation, and structuring processes, ensuring accuracy and consistency of time-sensitive reports
- Streamlined reporting procedures under high-pressure operational conditions by improving data quality and standardisation across multiple information sources
- Demonstrated rapid adaptability by learning operational reporting protocols and contributing to process improvements in a dynamic, mission-driven environment

Data Science Intern

Mar, 2023 - Nov, 2023

Neighbourlytics, Melbourne, Australia

- Built a geospatial analytics pipeline on Google BigQuery processing ~2M+ mobile device pings for large-scale spatial analysis and behavioural segmentation
- Designed and maintained spatial data schemas and metadata documentation to ensure data consistency, feature clarity, and reproducibility across analytical workflows
- Performed data preprocessing, validation, and feature standardisation to improve reliability of downstream clustering and analytical outputs
- Evaluated clustering methods (DBSCAN, Spectral Clustering) within a research team, supporting spatial modelling decisions and scalability trade-offs
- Developed a multi-stage spatial clustering framework, reducing GPS noise by ~35–45% and improving stability of behavioural segmentation outputs

RESEARCH & PROJECTS

Technical Mentor

Apr, 2026 - May, 2026

SNL-ICARUS (Cybersecurity Training & CTF Platform), Melbourne, Australia

- Guided design and development of a CTF-based cybersecurity training platform, structuring end-to-end challenge environments including web exploitation, authentication bypass, and data extraction scenarios
- Mentored participants through hands-on labs and guided problem-solving sessions, improving applied understanding of cybersecurity concepts and analytical thinking

Machine Learning Researcher

Jan, 2026 - Apr, 2026

Interpretable Machine Learning for Early Sepsis Detection, Melbourne, Australia

- Developed an interactive dashboard to visualise patient risk trajectories and feature-level impact, enabling interpretable analysis and clinician-facing insights
- Analysed MIMIC-IV ICU datasets (~2,000 admissions) to identify early sepsis risk patterns, transforming clinical time-series data into structured datasets for modelling and analysis
- Engineered analytical pipelines to support interpretable machine learning workflows for healthcare risk prediction

Machine Learning Engineer

Oct, 2025 - Jan, 2026

Cloud-based Social Sentiment Analytics Platform, Melbourne, Australia

- Analysed large-scale social media datasets (270K+ Twitter and Mastodon posts) to identify sentiment trends and behavioural patterns across platforms
- Designed NLP-based sentiment analysis workflows to extract insights on public opinion trends and temporal behaviour shifts
- Built automated data processing pipelines for ingestion, cleaning, and transformation of unstructured text data, enabling structured reporting and time-series analysis of sentiment trends

EDUCATION

Master of Data Science

Mar, 2023 - Nov, 2025

The University of Melbourne

- Focused on machine learning, statistical modelling, and scalable systems including cloud and distributed computing

Bachelor of Science (Data Science)

Mar, 2020 - Nov, 2022

The University of Melbourne

- Built strong foundations in algorithms, probability, statistics, and database systems

TECHNICAL SKILLS

- LLM & AI: RAG Systems, LLM Applications, Prompt Engineering, Few-shot Learning, LLM Evaluation, NLP
- Machine Learning & Modelling: PyTorch, TensorFlow, Scikit-learn, Statistical Modelling, Time Series Analysis, Feature Engineering
- Cloud & MLOps: AWS, Azure, Google Cloud Platform (BigQuery), Cloudflare, Docker, CI/CD, Serverless Architecture, Model Deployment & Serving
- Data Engineering: Apache Spark, BigQuery, Snowflake, ETL Pipelines, MySQL, CouchDB, Data Warehousing
- Data Visualization & BI: Tableau, Power BI
- Programming: Python, SQL, R, Java, JavaScript, TypeScript
- Certifications: CFA (Candidate), AWS Cloud Practitioner, ADsP (Data Analytics Semi-Professional)

REFEREES

John Kim

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